

American Elm

Ulmus americana

Species background: Native to eastern North America; once the single most-planted street tree in the United States, forming iconic tree-lined 'elm tunnels' over countless American streets before the mid-20th century. Fold a leaf in half at its base -- the two sides won't line up, because American elm leaves have distinctly uneven bases, a reliable ID trait year-round.

This packet contains one full lesson for each grade band: K-2, 3-5, 6-8, and 9-12. Each includes a learning objective, standards connection, materials, teacher background, a step-by-step procedure, discussion questions, an assessment, and an extension activity.

American Elm: Find American elm's uneven leaf base by folding a leaf.

Standards connection: NGSS K-LS1-1; National Core Arts Standards VA:Cr1 (K-2)

Materials: Coloring sheet, crayons.

Teacher background

Fold the leaf at the stem -- do the two sides match up? Long ago, trees like this covered whole streets like a green tunnel.

Procedure

- 1 Engage: Ask students to fold a piece of paper in half and check if the two sides match.
- 2 Explore: Fold a real or pictured elm leaf and check the two sides.
- 3 Explain: Explain that this uneven base is a special clue just for elm leaves.
- 4 Art: Fold a leaf at the stem and see if the two sides match.
- 5 Evaluate: Student can describe the leaf's two sides as uneven or 'not matching.'

Discussion questions

What other things come in two halves that don't quite match?

Assessment

Correctly folds and describes the uneven leaf base.

Extension

Try the fold test on a different tree's leaf and see if it matches evenly.

American Elm: Describe how elm-lined streets once looked and why that changed.

Standards connection: NGSS 3-LS4-4; National Core Arts Standards VA:Cr2 (3-5)

Materials: Coloring sheet, colored pencils.

Teacher background

Old American elms used to form green 'tunnels' over city streets before a disease hurt many of them.

Procedure

- 1 Engage: Ask students to imagine a street covered entirely by arching tree branches like a tunnel.
- 2 Explore: Look at historic photos of elm-lined streets if available.
- 3 Explain: Explain that a disease spread by beetles killed most of these trees last century.
- 4 Art: Draw the tall, arching branches of an elm forming a tunnel shape, like a cathedral made of branches.
- 5 Evaluate: Student can describe in one sentence what happened to America's street elms.

Discussion questions

How do you think a street looked and felt different after losing its elm tunnel?

Assessment

Correct one-sentence summary of Dutch elm disease's impact.

Extension

Find an old photo of your own neighborhood and see if any elms are visible.

American Elm: Explain monoculture risk using Dutch elm disease as a historical case study.

Standards connection: NGSS MS-LS2-1; National Core Arts Standards VA:Cr2/Re7 (6-8)

Materials: Coloring sheet, research materials on historic elm streets.

Teacher background

Dutch elm disease, spread by bark beetles, nearly wiped out American elm as a street tree in the 1900s -- a real case for why cities now plant many different species instead of one.

Procedure

- 1 Engage: Why might planting only one species along an entire street be risky, even if that species looks great?
- 2 Explore: Research how Dutch elm disease spread and how quickly American elm populations declined.
- 3 Explain: Introduce 'monoculture risk' as the lesson learned from this historical case.
- 4 Art: Research historic photos of elm-lined American streets before Dutch elm disease, then illustrate a 'before and after' comparison of a tree-tunnel street.
- 5 Evaluate: Write a paragraph explaining monoculture risk using American elm as the historical example.

Discussion questions

What modern street-tree planting practices reflect lessons learned from the elm disaster?

Assessment

Paragraph correctly explains monoculture risk with specific historical detail.

Extension

Research how many different tree species are planted on your own block or a nearby street.

American Elm: Evaluate species-diversity policy in modern urban forestry using the elm collapse as precedent.

Standards connection: NGSS HS-LS2-6; National Core Arts Standards Proficient-level Creating/Responding (9-12)

Materials: Coloring sheet, research sources, design materials for a poster or illustration.

Teacher background

Dutch elm disease spreads via bark beetles and root grafts between neighboring trees of the same species; research how modern urban forestry departments use species-diversity targets (for example, no more than 10% of one species) to prevent a repeat of the 20th-century elm collapse, and evaluate this street's own species diversity.

Procedure

- 1 Engage: What percentage of one species would you consider 'too much' for a city to plant, and why?
- 2 Explore: Research a real municipal species-diversity target or policy for street tree planting.
- 3 Explain: Evaluate whether the diversity you observe on your own walk reflects that kind of policy.
- 4 Art: Design a public-awareness illustration or poster about the importance of species diversity in urban forestry, using this tree's near-loss to Dutch elm disease as the central case study.
- 5 Evaluate: Submit the species-diversity evaluation and the awareness poster.

Discussion questions

Is a strict species-diversity cap always the right policy, or could there be tradeoffs (cost, availability, climate suitability) that argue against it in some cases?

Assessment

Rubric covering policy research accuracy and poster communication effectiveness.

Extension

Compare American elm's monoculture collapse to green ash's ongoing emerald ash borer crisis -- has the lesson actually been applied?

SOURCES & METHODOLOGY

Background research drawn from TreeWalk NYC's own species research. Lesson structure (objective, background, materials, procedure, assessment, extension) follows the format used by real, freely published environmental-education lessons, including the National Park Service's "Looking at Leaves" 5E lesson plan (Acadia Outdoor Classroom Collaborative), Penn State Extension's "Sorting and Classifying Leaves," and Project Learning Tree's leaf-activity guidance -- adapted and written fresh for this species, not copied from those sources. This is an educational pilot; standards references indicate topic alignment, not formal certification.